



A Natural Ester Based Dielectric Fluid

Retrofilling of Transformers with bioTRANSOL HF Natural Ester Fluid: General considerations

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1.1 Retrofilling of transformers with Biotransol HF Natural Ester fluid

bioTRANSOL is Savita make India's first indigenous Natural Ester fluid. It is tested as per relevant IEC & IS standards in national & international laboratories and carefully manufactured in Savita's state of the art facility.

It is FM Global & UL approved fluid which certifies it as less flammable fluid. It has been tested for its biodegradability and acute aquatic toxicity and shown satisfying results.

It is an ideal fluid for use in sealed Distribution & Power transformers which offers several benefits to the end user over conventional mineral oil and adds value to the equipment.

Retrofilling of existing mineral oil filled Transformer with Natural Ester fluid offers following benefits to the end user,

- Offering better fire safety,
- Slows down the further degradation of insulation paper condition and hence enhance the transformer life.
- Additional increase in the transformer life shall differ the capex cost of replacing the transformer at the end of life.
- Maintenance shall be less as natural ester fluid offers no sludge formation as in case of mineral oil.

1.2 Pre requisites

Following are the pre requisites to be confirmed/obtained before the transformer is taken for the retrofilling,

1. Checking the Transformer insulation condition: For the transformers with considerable operational life, it is important to check the condition of the insulation paper. Furan analysis shall be performed on the oil sample from which the DP value of the paper can be obtained. Generally DP value of 400 and higher can be considered to be viable to proceed further for Power Transformer.

2. Original Transformer manufacturer's operation and maintenance manual to be obtained.

3. WTI & OTI log should be available to assess the efficiency of existing cooling system and loading pattern.

The retrofilling of transformers can be undertaken in two ways:

- A. On site Retrofilling: In case of in-service medium range power transformers, the retrofilling can be done at site based on Transformer condition assessment. Taking these transformers out from service and transporting to and from factory is not only costly and time consuming, it is risky also as it may result in damage to coils, insulation etc. However if feasible, it is recommended to take the Transformer to the nearest workshop for retrofilling activities.

As this range of Power transformers are generally designed with aircell in the conservator and conservator protection relay (CPR), retrofilling involves only replacement of mineral oil with ester fluids and changing gaskets from cork rubber to Nitrile bonded rubber. Generally other modifications are not feasible.

If some old Power Transformers which are not provided with Aircell in the conservator and air cell protection relay; Design modifications has to carried out as per procedure mentioned in serial no. 1.3 below.

- B. Offsite Retrofilling: In case of free breathing type transformers, retrofilling, preferably should be done in a factory. Such transformers are required to be modified from free breathing type to sealed type.
- C. Refer our handling and storage guidelines before commencement of retrofilling.

1.3 Design Modifications for Retrofilling:

i. Tank consideration: The non-sealed type the transformers being retrofilled must be modified suitably to ensure that ester fluid does not come in contact with outside atmosphere. Open breathing transformer to be modified by providing an air cell in the conservator and conservator protection replay (CPR).

ii. Gaskets replacement: As ester filled transformers operate at higher temperatures, it is recommended to replace gaskets by nitrile or better materials. If nitrile is being used then proper retainers must be provided to prevent flaring up of the same under temperature and pressure during service.

While handling natural ester fluids, following must be noted as esters behave differently in comparison to mineral oil:

- a. Proper care has to be taken during handling of ester fluid to minimize contact with atmosphere.

- b. The traces of ester fluid film on the surfaces leave a hard layer if not cleaned immediately hence ensure proper cleaning of surfaces.

1.4 Procedure for Retrofilling of the Transformer at site:

1. Inspect the transformer thoroughly for visible leakages, bulging of tank wall etc.
2. In order to make the existing free breathing Transformer sealed type; existing conservator to be modified to make it suitable for Ester fluid. Aircell mounting arrangement is to be provided inside the conservator along with MOG & Conservator protection relay.
3. Once the conservator is ready with Aircell & MOG, check the MOG operation with respect to aircell.
4. Following materials/equipments needs to be kept ready before the start of retrofilling procedure;
 - a. Modified conservator as above,
 - b. Set of Nitrile Gasket,
 - c. Adequate quantity of new unused bioTRANSOL Natural Ester fluid,
 - d. Vacuum filtration system
5. In order to ensure that natural ester retains its high fire point of > 300 deg c, residual mineral oil should not be more than 7% in the retrofilled transformers. Therefore, sufficient time must be allowed to drip off maximum oil from the core, coil and insulation during draining off stage of mineral oil. Residual oil can be removed off and stored in storage tank/barrels for further disposal as per regulations. Drip off time has to be decided based on the transformer rating. Higher rating transformers may need longer drip-off time compared to lower rating transformers.
6. The leak test is recommended with dry nitrogen. Compressed air should NOT be used for this purpose.
7. Gaskets to be replaced with Nitrile Rubber gaskets.
8. The inside of the transformer should be flushed thoroughly with natural ester to remove traces of mineral oil from the Transformer. The quantity of ester oil used for this purpose may be 10% of the total oil volume. The flushed oil should be drained off fully and disposed off suitably.
9. The filtration machine should be rinsed with Natural Ester to remove any traces of Mineral Oil to avoid contamination in case a common filtration machine is used.
10. It is advisable and recommended to fill the Natural Ester under full vacuum using a machine having 0.5 micron filter medium.
11. The filling of the Natural Ester should be done by maintaining oil temperature up to 60 to 70 Deg. C. and adopting standard oil filling practices.

12. An adequate settling time must be maintained before offering the transformers for final testing for proper impregnation and to avoid the air trap.
13. Natural Ester fluid sample to be taken soon after filling as above and sent to the laboratory for testing for future reference.
14. After completion of entire work, all the mandatory electrical tests should be performed on the Transformer before charging.
15. The insulation parameters like insulation resistance, dissipation factor etc test lower as compared to mineral oil, and suitable benchmarks should be generated for future.
16. As it is not feasible to change old insulation, cooling design, dielectric design etc. in existing mineral oil filled transformer; we may need to consider higher temperature rise than mineral oil after retro-filling due to higher viscosity of natural ester fluid.
17. In case of retrofilling, if the existing paper insulation is not a thermally upgraded paper, overloading possibilities for transformers are limited.
18. As the esters have high viscosity and a pour point of -18 to -21°C , retrofilling is recommended with cold start procedure in case the ambient temperatures is less than -10°C .
19. Transformer should be energized by following standard Transformer commissioning practices.
20. It is recommended to test oil parameters twice in the first year (6 monthly) and then once in every year.

1.5 Additional points to be considered (For the transformers which are undergoing repair)

Following points should be considered during retrofilling of the transformers which have undergone repairing of core coil assembly and in the open condition at the shop floor:

- a. As ester has a tendency to get oxidized, Tank cover should be put in place as early as possible. In case this gets delayed, one pass filtration should be done of complete fluid inside the tank.
- b. If the core coil assembly soaked in ester has to be kept outside, in order to avoid the film formation of ester fluid in the surface of core coil assembly; it is advisable to flush it thoroughly with mineral oil.
- c. All the electrical contact surfaces - like LV bus bars, Jumpers, Tap switch contacts, HV and LV terminals etc., of ester soaked core-coil assembly must be thoroughly cleaned with Acetone or equivalent solvent. This is required to be done to remove the oxidized ester coating from the contacts, else, electrical resistance values will be erroneous.

Additional Note regarding accessories of Transformer:

- i. Radiators: Radiators should be thoroughly checked for leakages etc and attended before commencing the work. Working of cooling fans, pump etc. should be checked for efficiency.
- ii. OLTC: If user wish to replace the oil in OLTC then respective OLTC manufacturer to be contacted.
- iii. Bushings: Existing bushings with mineral oil to be continued. However, if user wish to replace the bushings, bushings suitable for ester application to be selected.

Note: The above procedure is suggested based on general and common manufacturing practices adopted by industry and may vary with different regions.

1.6 Reference Standards:

Sr.	Standard	Standard Number
1	Outdoor type Oil Immersed Distribution Transformers Upto and including 2500 kVA, 33 KV Specification (Mineral Oil Immersed)	IS 1180 Part 1: (2014)
2	Outdoor type Oil Immersed Distribution Transformers Upto and including 2500 kVA, 33 KV Specification with Ester Fluids	IS 1180 Part 2
3	Indian Standard for Power Transformer	IS 2026
4	Fluids for electrotechnical applications – Unsused Natural Ester for Transformers and similar Electrical Equipments	IEC:62770:2013 & IS:16659
5	Liquid-Immersed Power Transformers Using High-Temperature Insulation materials	IS 2026 Part 14 & IEC 60076-14
6	IEEE Standard for the design, testing & application Of liquid immersed Distribution, Power & regulating Transformers Using High Temperature Insulation Systems & Operating at elevated Temperatures	IEEE C57.154 – 201
7	IEEE Guide for interpretation of Gases generated in Natural and Synthetic Ester – immersed Transformers.	IEEE C57.155
8	Guide for Interpretation for gases generated in Natural Ester & Synthetic Ester immersed Transformers	IS 16785
9	IEEE Guide for acceptance and maintenance of Natural Ester fluid in Transformers	IEEE C57.147.2008
10	Natural esters – Guidelines for maintenance and use in electrical equipment	IEC 62975:2021
11	Use and Maintenance of Transformer: Guide for sampling Gases and oil from oil filled Electrical equipment for the analysis of free and Dissolved Gases	IEC 60567(1992)
12	Power installations exceeding 1 kV a.c. – Part 1: Common rules”	IEC 61936-1
13	Biodegradation	OECD 301 B
14	Fittings and accessories for power transformers	IS:3639
15	Code of practice for selection, installation & Maintenance of transformer	IS:10028
16	Guide for loading of oil immersed transformers.	IS:6600

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